

Noel Joe Basil

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Summary

Forward-deployed data engineer with 8+ years of experience delivering end-to-end analytical platforms across health-care and industrial domains. Deep expertise across the full Palantir Foundry stack—data integration, Ontology design, Python Functions, OSDK React applications, and AIP/LLM-powered workflows. Track record of owning solutions from business problem through to production deployment, working directly with senior stakeholders in highly regulated environments.

Deployment Highlight: London Vaccination Surveillance Platform

Sole technical owner of a national vaccination surveillance reporting system serving NHS England's London region across seasonal flu and Covid campaigns (30 weekly reporting cycles/year).

Problem: Legacy process required analysts to manually refresh a Contour analysis, export charts to PowerPoint, and hand-write surveillance commentary each week—with recurring seasonal breakage requiring logic changes every campaign.

Solution deployed in two iterations:

- **V1:** PySpark pipelines → Ontology (5 object types) → Quiver dashboards (Vega) → Notepad narrative + PDF export
- **V2:** Python Functions (16 endpoints) → OSDK React app → LLM-powered narrative generation → automated PDF reports

Impact: Season-agnostic architecture eliminated recurring seasonal maintenance. Automated narrative generation across 30 weekly cycles. Self-service reporting replaced analyst-mediated delivery. One-click PDF export replaced multi-tool manual workflow.

Technical Skills

- **Palantir Foundry:** Code Repositories, Pipeline Builder, Ontology (Object Types, Links, Actions), Python Functions, OSDK (React/TypeScript), AIP (LLM integration), Workshop, Quiver, Vega, Contour, Notepad
- **Data Engineering:** PySpark, Python, SQL, TypeScript, ETL/ELT pipelines, data modelling, incremental transforms, data quality frameworks
- **AI/ML:** LLM integration (Foundry AIP), prompt engineering, automated narrative generation, predictive modelling
- **Cloud & DevOps:** Azure, Databricks, GitHub Actions, Terraform, CI/CD, monitoring (Application Insights)
- **Visualisation & BI:** Power BI, Tableau, Vega, Quiver, Contour
- **Governance:** PDP/access control design, Information Governance, markings, data security in regulated environments

Professional Experience

Analyst — Data Engineering / Analytics Engineering

NHS England

2025 – Present

- Deployed end-to-end Foundry solutions for London regional vaccination surveillance, owning full stack from data integration through Ontology design, Functions, and front-end applications.
- Designed and built PySpark ETL pipelines across 5 Code Repositories, modelling complex multi-source healthcare datasets into a governed, season-agnostic Ontology layer.
- Developed 16 Python Ontology Functions as a data access layer, computing real-time statistics including uptake summaries, cohort breakdowns, equity analyses, and operational metrics.
- Built a full-stack OSDK React application with interactive charts and automated PDF generation, replacing manual Quiver/Notepad workflows.
- Integrated LLM-powered narrative generation via Foundry AIP, automatically producing surveillance commentary across 11 dashboard sections with domain-specific prompt engineering.
- Architected PDP-based access layers and implemented Information Governance controls aligned with NHS data policies, enabling secure multi-stakeholder data consumption.

- Delivered custom Quiver applications with Vega visualisations and Notepad-based structured outputs for senior stakeholder consumption (V1 of reporting platform).
- Developed statistical models (logistic regression, stratified analyses) on 400k+ screening episodes identifying inequality drivers in breast screening uptake, informing regional commissioning.
- Built Power BI dashboards for maternity, school-age vaccination, and COVER reporting programmes.
- Collaborated directly with clinicians, programme leads, and regional analysts to translate complex requirements into deployed solutions.

Analytical Consultant

Merkle UK (Oil & Gas Client — Offshore Operations)

2021 – 2025

- Deployed production-grade analytics and optimisation systems leveraging IIoT sensor data across offshore platforms, working directly with operations teams to translate operational problems into deployed solutions.
- Built predictive models for hydrate formation and heat exchanger fouling, increasing maintenance efficiency by 50%.
- Deployed generator optimisation system for shutdown operations, reducing fuel consumption and emissions by 50%.
- Developed real-time compressor optimisation logic evaluating flow rates against safety constraints, directly improving operational efficiency.
- Re-architected legacy PySpark workflows into production-stable Python pipelines, eliminating recurring failures and reducing execution time.
- Designed CI/CD pipelines (GitHub Actions) enabling automated testing, security scanning, and deployment to Azure, introducing engineering rigour into an analytics function.
- Delivered enterprise Power BI dashboards replacing manual reporting, saving 2 hours daily across offshore teams. Implemented deployment pipelines for version-controlled BI.
- Worked across distributed teams spanning offshore operations and onshore analytics, adapting communication to technical and non-technical stakeholders.

Decision Scientist

Mu Sigma

2016 – 2019

- Delivered Tableau dashboards and automated SQL pipelines supporting enterprise-scale reporting for Fortune 500 clients.
- Translated business requirements into scalable analytical solutions, working across cross-functional teams.

Education

MSc Business Analytics (Distinction) — The University of Edinburgh Business School

2020 – 2021

B.Tech. Electronics and Communication Engineering — The University of Kerala

2012 – 2016

Training & Professional Development

- Palantir Foundry Training (Data Integration, Pipeline Development, Ontology, Platform Capabilities) — Palantir Learning Platform
- Machine Learning Zoomcamp — DataTalksClub (Containerisation, MLOps, deployment pipelines)